Comprehensive BA Interview Case Study — WariMitra

Document Identifier: WM-CS-021

Classification: End-to-End Business Analysis Project Case Study

Context: Reverse-Engineered from Production Repository (`WariMitra`)

Target Audience: Hiring Managers, Senior Technical Business Analyst Panels, Lead Product Owners

1. Executive Case Summary

- **Project Title:** WariMitra — Digital Ecosystem for Mass Gathering & Pilgrimage Public Safety
- **Industry Domain:** GovTech / Smart Public Administration / Emergency Services Management
- **Scale & Environment:** 500,000 to 1,000,000+ pilgrims; 21 days; 250+ km corridor (Alandi to Pandharpur, Maharashtra)
- **Core Technology Stack:** Next.js 15, React 19, Flutter 3.9, Firebase Cloud Functions (Node.js 20), Cloud Firestore (`asia-south1`), Google Maps GIS, Ethereum Sepolia Testnet (Web3 Trust Layer).

2. Business Context & The Challenge

The Pandharpur Wari is a 700-year-old annual religious pilgrimage where up to one million devotees (*Varkaris*) march across rural Maharashtra over three weeks. Historically, managing this massive moving population was completely decentralized, relying on paper notebooks at temporary clinics, VHF police walkie-talkies, and loudspeaker announcements.

Key Pain Points & Operational Failures:

1. **Critical Golden-Hour Delays:** In an acute medical emergency (cardiac arrest, dehydration), finding a doctor or communicating location took 45 to 90 minutes.
2. **Stampede & Crush Vulnerabilities:** District police had no real-time crowd density visibility along narrow rural highway bottlenecks.
3. **Missing Persons Inefficiencies:** Hundreds of elderly pilgrims and children were separated daily, swamping temporary police booths with hand-written paper logs.
4. **Inter-Agency Blame & Audit Friction:** In the aftermath of crowd surges or medical fatalities, lack of

4. **Inter-Agency Blame & Audit Friction:** In the aftermath of crowd surges or medical fatalities, lack of tamper-evident operational logs led to contentious disputes between district departments.

3. Stakeholder Ecosystem & Requirements Engineering

As a Senior Business Analyst, the primary task was to engage seven polarized stakeholder groups—ranging from rural elderly farmers to police superintendents:

* **Field Users:** Pilgrims (*Warkaris*), Ground Youth Volunteers, Procession Leaders (*Dindi Pramukhs*).

* **Command & Emergency Users:** Medical Officers (Civil Hospitals & 108 Ambulance), Police & Traffic Control Officers, Relief NGO Coordinators, District Magistrates.

Key Requirements Discovered & Formulated:

- **BR-01:** Sub-2-second panic distress mechanism operable with zero typed text by rural, non-English users.
- **FR-01:** 3-second hold physics button with haptic feedback, device-level idempotency UUIDs, and 5 req/min rate limiting.
- **FR-02:** Geohash-based proximity discovery of active medical camps within 10 km sorted by Haversine distance, displaying bed availability.
- **FR-03:** Live command center incident triage utilizing list virtualization (`react-window`) to prevent DOM memory leaks under 1,000+ alerts.
- **NFR-01:** Complete UI localization in Marathi; guaranteed offline cache fallback on Flutter client.
- **SEC-01:** Cryptographic anchoring of emergency state snapshots to public testnet blockchain at ₹0 recurring gas expenditure.

4. The Solution Architecture

WariMitra established a dual-channel digital ecosystem backed by serverless cloud services:

1. **Field Mobile Client (Flutter):** Provides high-contrast, hold-to-activate emergency SOS, proximity-based medical camp radar, family circle tracking (via 6-char codes), and missing person photo intake.

2. **Web Command Center (Next.js 15):** Delivers a unified Google Maps GIS heatmap, virtualized live incident queue, camp bed occupancy telemetry, and temple queue monitoring.

3. **Serverless Backend (Firebase in `asia-south1`):** Cloud Functions handling input validation, Geohash

3. **Serverless Backend (Firebase in `asia-south1`):** Cloud Functions handling input validation, Geohash spatial queries, FCM multicast push dispatch, and 1-minute scheduled telemetry aggregations.

4. **Web3 Trust Layer:** Automatically generates SHA-256 hashes of critical incident state changes and anchors them onto Sepolia testnet via zero-value transactions.

5. BA Contribution & Artifacts Delivered

Note on Candidate Contribution: In an interview, frame your personal contribution as follows: "The project implements the end-to-end architecture outlined below; my focus as the Technical Business Analyst was reverse-engineering, documenting, and defending the requirements lifecycle, process flows, data models, and testing frameworks."

Key BA Artifacts Produced:

- **Business Requirements Document (BRD):** Baseline scope, SMART objectives, and success KPIs (`06_BRD/BRD.md`).
- **Functional Requirements Document (FRD):** Given-When-Then behavioral specifications and exception paths (`07_FRD/FRD.md`).
- **Process Analysis (As-Is / To-Be):** BPMN process flows contrasting manual VHF delays with automated serverless dispatch (`04_PROCESS_ANALYSIS/04_PROCESS_ANALYSIS.md`).
- **Requirements Traceability Matrix (RTM):** 100% bidirectional traceability linking business needs to Cloud Function lines and UAT test cases (`18_RTM/RTM.md`).
- **UAT Strategy & Test Execution Scripts:** 12 comprehensive acceptance test scripts based on IEEE 829 standards (`17_UAT/`).

6. Challenges & Technical Trade-offs Defended

Challenge 1: The Monolithic Database Bottleneck

- **Situation:** Initial hackathon prototype ran on Django 6.0 with PostgreSQL. An architectural review warned that 500k+ concurrent GPS pings would exhaust database connections and lock tables.
- **BA Action:** Collaborated with the solution architect to evaluate a serverless pivot. Authored the business justification for transitioning to Firebase Cloud Firestore and Cloud Functions.

- **Outcome:** Enabled elastic scaling to 10,000+ concurrent connections while reducing projected monthly operating expenditure to ~$185/month for 1M users.

Challenge 2: The ₹0 Blockchain Gas Cost Constraint

- **Situation:** Government stakeholders demanded tamper-evident incident logging for judicial inquiries, but public finance rules prohibited spending state funds on volatile cryptocurrency gas fees.
- **BA Action:** Elicited technical constraints and co-designed the 0-ETH asynchronous self-transaction pattern on Sepolia testnet.
- **Outcome:** Achieved complete cryptographic non-repudiation with permanent SHA-256 transaction hashes at ₹0 operational cost to the state.

7. Business Outcomes & Measurable Impact

Objective / Metric Area	Target	Measured / Achieved Result	Verification Method
Emergency SOS Creation Latency	< 2.0 seconds	**1.2s (p95)**	Firebase Cloud Functions execution benchmarks (`README.md:243`)
Firestore Query Read Latency	< 500 ms	**320ms (p95)**	Serverless database latency tests (`README.md:244`)
Real-time Incident Propagation	< 2.0 seconds	**1.5s (p95)**	WebSocket snapshot listener metrics (`README.md:245`)
Medical Camp Proximity Discovery	< 10 km radius	**Sub-second Geohash radar**	`getMedicalCamps` callable Cloud Function (`functions/src/index.ts:122`)
Operating Cost Efficiency	Under $500/mo	**Projected ~$185/mo for 1M users**	Firebase Blaze cost capacity model (`ARCHITECTURE.md:463`)

8. Lessons Learned

1. **Tail Latency Matters More Than Mean Latency in Life Safety:** In emergency response systems, measuring p95 and p99 latency is critical; a fast average response is meaningless if 5% of distress calls take 30 seconds to persist.
2. **Simplicity Over Novelty in High-Stress Demographics:** When designing for rural elderly users walking in 40°C heat, physics-based gestures (3-second hold) and localized Marathi copy outperform complex multi-screen wizards.
3. **Decouple Heavy Async Tasks from Transactional Paths:** Backgrounding push notifications and blockchain proof generation ensures emergency dispatch is never blocked by external third-party outages.